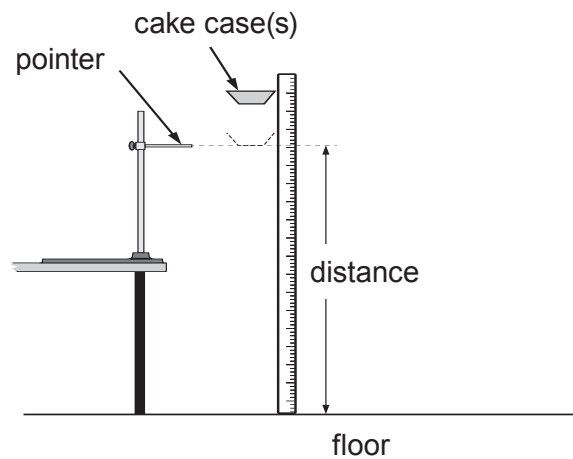


Examiner
only

- 5.** Students investigate the terminal speed of falling paper cake cases.



- (a) (i) State what is meant by terminal speed.

[1]

- (ii) Describe how you would carry out the investigation to find out how the mass of falling paper cake cases affects their terminal speed. [6 QER]

[6 QER]



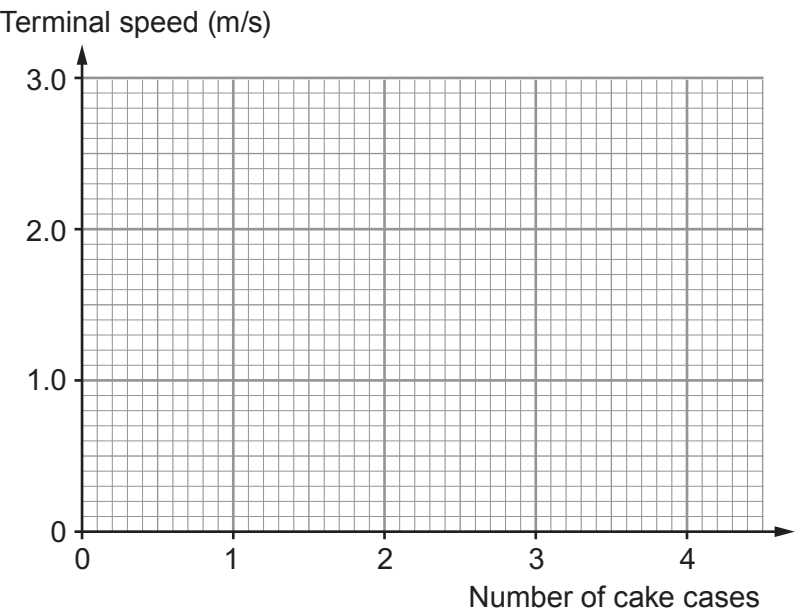
Examiner
only

(b) The table shows results from the investigation.

Number of cake cases	Distance (m)	Mean time taken for cake cases to fall 1.50 m (s)	Terminal speed (m/s)
0			0
1	1.50	0.90	1.7
2	1.50	0.68	2.2
3	1.50	0.60	
4	1.50	0.56	2.7

(i) **Complete the table** using an equation from page 2 to find the value of the missing terminal speed. *Space for workings.* [2]

(ii) Plot the data in the table on the grid below and draw a suitable line. [3]



(iii) One student suggests that doubling the number of cake cases reduces the **time** to fall by a half. Explain whether the results support this suggestion. [3]

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